

Public Reproducibility Metadata
for
A Triadic Solution to the General N-Body Problem

Allan Edward
Independent Researcher

July 6, 2026

Abstract

This companion metadata file specifies the public information needed to understand and reproduce the claim made in *A Triadic Solution to the General N-Body Problem*. It records the theorem object, triadic definitions, locally generated witness protocol, normalized-time convention, no-loss quotient rule, public validation source, query descriptors, validation protocol, constants, claim boundaries, and reproduction path. The file deliberately avoids private development archives, private working files, and project-history material. Reproduction is intended to proceed from the preprint, the definitions in this metadata file, and public NASA/JPL Solar System Dynamics query descriptors. Public reference trajectories are read-only validation data and are not admission witnesses.

Contents

1 Purpose and Scope	3
2 Claim Summary	3
3 Triadic Mathematical Foundation	3
3.1 Related Public Triadic Preprints	5
4 Author and Citation Metadata	5
5 Triadic Definitions	5
6 Kernel and Coefficient Form	6
7 Local Witness and Prediction Protocol	7
8 Axioms and Formal Proof Inventory	8

9 Proof Dependency Map	9
10 Public Validation Source	10
11 Public Query Descriptor Classes	10
12 Numerical Validation Protocol	11
12.1 Jacobi Recalculation	11
12.2 Propagation Check	11
13 Expected Public Validation Results	12
14 Independent Proof Reproduction Procedure	12
15 Claim Boundaries	13
16 Public Bibliographic References	14
A Minimal Public Reproduction Checklist	14
B Public Metadata Integrity Statement	14

1 Purpose and Scope

This metadata file is a public reproducibility companion to the preprint titled *A Triadic Solution to the General N-Body Problem*. Its purpose is to make the claim, proof dependencies, public validation source, and reproduction procedure clear without disclosing private development materials.

This metadata file is not a replacement for the proof. The proof is contained in the preprint. This file supports independent reconstruction by recording definitions, constants, public data references, validation protocols, and claim boundaries.

Public Scope

The public package consists of:

1. the preprint manuscript;
2. this public reproducibility metadata file;
3. the equations, definitions, query descriptors, and validation procedures stated here.

Excluded Materials

Private working archives, intermediate development notes, and author-side derivation history are not required for reproduction and are not part of the public record. The public validation layer is regenerated from NASA/JPL public interfaces, not from private copies of downloaded material.

2 Claim Summary

The public claim is the following.

The author defines a constructive triadic closed-form kernel-series closure operator for arbitrary finite N body-family dynamics under certified admissibility, modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy.

The title uses the phrase “general N-body problem” in the sense formalized by the theorem: arbitrary finite N , directed body-reference lanes, triadic closure, constructive kernel-series representation, and collision closure modulo lawful branch policy. The claim is not a finite elementary coordinate formula and not a machine-checked formal proof.

3 Triadic Mathematical Foundation

The triad is a formal operator structure, not an arbitrary name. The public construction uses the ordered packet

$$P = (\text{CIF}, \text{QV}, \text{RFL}, M_{\text{rec}}, K_f),$$

where CIF is the source envelope, QV is witnessed lawful action, RFL is stabilized return lineage, M_{rec} is return memory, and K_f is triadic kernel action.

The First Action is

$$\text{QV}(\text{CIF}) \longrightarrow \text{RFL}.$$

The recursive return cycle is

$$\text{CIF}_s \longrightarrow \text{QV}_s \longrightarrow \text{RFL}_s \longrightarrow M_{\text{rec}} \longrightarrow \text{CIF}_{s+1}.$$

This is the mathematical basis for requiring every admitted body-reference lane, route, collision event, and closure branch to carry a full CIF/QV/RFL/ M_{rec} / K_f witness.

The basic triadic kernel is

$$K_f(t) = \sum_{j=0}^{\infty} \delta^{-j} e^{-\alpha j t} f_j(t).$$

The present N-body manuscript lifts this kernel from a single level function $f_j(t)$ to route-indexed finite- N body-family functions $\Phi_{j,[r],N}(t)$.

Time and recurrence convention

The recurrence index s and kernel depth j encode ordered inheritance of stabilized memory. They do not introduce a second independent physical clock. The symbol t denotes normalized dynamical time throughout the N-body operator. If dimensional time t_{phys} is used, a fixed domain nondimensionalization $t = \mathcal{N}_N(t_{\text{phys}})$ must be declared before route generation and may not be fitted to validation data. This makes $\alpha j t$ dimensionless while preserving the same domain-time argument in the state, route coefficients, and route functions.

Witness meaning

A witness is a local admissibility certificate generated from the current admitted state, the governing differential residual, invariant and projection checks, collision policy, return memory, and kernel boundedness. A known public orbit is not an admission witness merely because it is already solved.

The constants are inherited as fixed structural constants from the author's public triadic preprints and are not fitted to the NASA/JPL validation layer. The constants recorded for reproduction are

$$\delta = 4.6692, \quad \alpha = 0.0256831, \quad \nu = 0.00420784,$$

$$\epsilon = 0.000108071, \quad \lambda_{\text{norm}} = 0.489442,$$

with packet weights

$$w_{\text{CIF}} = 0.005085, \quad w_{\text{QV}} = 0.984868, \quad w_{\text{RFL}} = 0.010047.$$

Readers seeking the broader derivation of the triad, the First Action, and the constant hierarchy should consult the author's prior public preprints listed in [Section 3.1](#).

3.1 Related Public Triadic Preprints

The following DOI records are provided as public references for the broader triadic framework and prior public applications. They are not private development files and are not required as hidden inputs to the present proof.

1. Allan Edward, public Zenodo record, DOI: [10.5281/ZENODO.20604240](https://doi.org/10.5281/ZENODO.20604240).
2. Allan Edward, public Zenodo record, DOI: [10.5281/ZENODO.20693393](https://doi.org/10.5281/ZENODO.20693393).
3. Allan Edward, public Zenodo record, DOI: [10.5281/ZENODO.20917761](https://doi.org/10.5281/ZENODO.20917761).

Central Structural Feature

A central feature of the construction is that adding bodies does not merely create additional obstruction. For a finite body set $\mathcal{B}_N$ with $|\mathcal{B}_N| = N$, the directed body-reference lane structure has

$$N(N - 1)$$

directed lanes. Adding one body increases the directed-lane structure by

$$(N + 1)N - N(N - 1) = 2N.$$

Those additional lanes can create additional lawful closure routes. In this framework, additional lawful closure routes are treated as solution capacity, not ambiguity.

4 Author and Citation Metadata

Field	Value
Author	Allan Edward
Affiliation	Independent Researcher
Primary manuscript	<i>A Triadic Solution to the General N-Body Problem</i>
Companion file	Public reproducibility metadata
Date	July 6, 2026
Subject area	Celestial mechanics; dynamical systems; constructive closure; N-body dynamics
Public validation source	NASA/JPL Solar System Dynamics Three-Body Periodic Orbits API

5 Triadic Definitions

The triadic terms are used as formal objects in the preprint.

Term	Public definition
CIF	Cosmic Infinite Field. Source and possibility envelope for admissible body-family configurations, route candidates, lane relations, and closure sources.
QV	Quantum Vacuum. Lawful collapse, compression, deletion, selection, and event-witness operator.
RFL	Recursive Feedback Loop. Stabilized emitted recurrence, return structure, and route-family memory.
M_{rec}	Return-memory map. The recurrence channel by which stabilized closure output is carried back into the next admissible source state.
K_f	Triadic kernel. The route-indexed closure kernel that weights depth, damping, route class, and triadic packet structure.
First Action	The primitive directed action $\text{QV}(\text{CIF}) \rightarrow \text{RFL}$.
Recurrence order	The ordered $\text{CIF}_s \rightarrow \text{QV}_s \rightarrow \text{RFL}_s \rightarrow M_{\text{rec}} \rightarrow \text{CIF}_{s+1}$ memory sequence. It organizes history but is not a second independent clock.
Normalized dynamical time t	The declared dimensionless domain-time argument used in state variables, route functions, coefficients, and kernel damping.
Local witness certificate	Locally generated differential-residual, invariant, projection, packet, branch, return-memory, and Φ -bound record used for route admission.
$L(i j)$	Directed body-reference lane from body i through reference body j , with $i \neq j$.
Gauge equivalence	Equivalence between closure representatives that produce the same lawful body-family closure.
Multi-route equivalence	Equivalence class in which distinct lawful closure routes are treated as route strength rather than obstruction.
Lawful collision branch policy	Rule selecting a post-collision branch only when conservation, reversibility, kernel continuity, projection consistency, and triadic witness distinguish it; otherwise the tied branch family is retained as lawful gauge symmetry.

6 Kernel and Coefficient Form

The constructive closure operator is expressed by a route-indexed triadic kernel series. A representative public form is

$$K_f, N(t) = \sum_{[r] \in \mathcal{R}_N / \sim} W_{[r], N}(t) \sum_{j=0}^{\infty} \delta^{-j} e^{-\alpha j t} \Phi_{j, [r], N}(t),$$

where $[r]$ is a lawful route class modulo gauge and multi-route equivalence, j indexes kernel depth, and $\Phi_{j, [r], N}$ is the bounded triadic packet function attached to the depth and route class. The symbol t is normalized dynamical time, not an additional fitted variable. Recurrence order and depth organize memory within the same time-parameterized domain construction.

A public coefficient form used for reproduction is

$$W_{[r], N}(t) = \frac{\chi_{\text{adm}}([r], N, t) S_{\text{CIF}}^{w_{\text{CIF}}} S_{\text{QV}}^{w_{\text{QV}}} S_{\text{RFL}}^{w_{\text{RFL}}} M_{\text{rec}} B G D_j}{Z_N}.$$

Here χ_{adm} is the admissibility indicator, S_{CIF} , S_{QV} , and S_{RFL} are triadic source, lawful-collapse, and recurrence scores, B is branch-policy certification, G is gauge-class certification, D_j is the kernel-depth damping factor, and Z_N normalizes the admissible route-class family.

Constants

Constant	Public value / role
δ	4.6692, geometric kernel-depth scaling constant.
α	0.0256831, fixed damping coefficient in the dimensionless exponent $e^{-\alpha j t}$ with normalized dynamical time t .
w_{CIF}	0.005085, triadic source-weight exponent.
w_{QV}	0.984868, triadic lawful-collapse-weight exponent.
w_{RFL}	0.010047, triadic recurrence-weight exponent.

The convergence argument uses $\delta > 1$, hence $\sum_{j \geq 0} \delta^{-j}$ is geometrically summable. Under certified boundedness of $\Phi_{j,[r],N}$ and admissible normalization over route classes, the kernel series is constructive under the stated admissibility conditions.

7 Local Witness and Prediction Protocol

For an admitted normalized-time state

$$\mathcal{S}_N(t_0) = (m_i, q_i(t_0), v_i(t_0))_{i=1}^N,$$

define on a noncollision chart

$$a_i(q) = G \sum_{k \neq i} m_k \frac{q_k - q_i}{\|q_k - q_i\|^3}, \quad r_{ij} = q_i - q_j, \quad \nu_{ij} = v_i - v_j.$$

For a candidate local route segment or jet γ , the lane residual is

$$\mathcal{R}_{ij}^\gamma(t) = \left(\begin{array}{c} \dot{r}_{ij}^\gamma(t) - \nu_{ij}^\gamma(t) \\ \dot{\nu}_{ij}^\gamma(t) - (a_i(q^\gamma(t)) - a_j(q^\gamma(t))) \end{array} \right).$$

Formal admission requires exact residual vanishing. A numerical reproduction may instead use a declared certified bound together with declared drift bounds for energy, total momentum, angular momentum, and center-of-mass relations. The certificate must also record projection consistency, packet continuity, collision/event-lift status, branch policy, return memory, and a common Φ bound.

Candidate paths may be proposed by Taylor-jet transport, collocation, interval propagation, implicit local solve, or another local constructor. The admission decision is method-independent: it is determined by the certificate. A pre-solved NASA/JPL orbit may be used for later validation but may not supply the admission witness.

The public prediction order is

$$\text{initial state} \rightarrow \text{lanes} \rightarrow \text{candidate jets} \rightarrow \text{certificates} \rightarrow \text{route classes} \rightarrow \text{kernel closure}.$$

8 Axioms and Formal Proof Inventory

The public proof is based on the following explicit axioms and proof objects. These are stated so that the theorem can be reconstructed without private notes or development history.

Axioms / Standing Structural Assumptions

Axiom	Public role
Triadic packet admissibility	Every admitted lane, route, collision event, and returned body-family state carries $(\text{CIF}, \text{QV}, \text{RFL}, M_{\text{rec}}, K_f)$.
First Action	The primitive operation is $\text{QV}(\text{CIF}) \rightarrow \text{RFL}$.
Witnessed admission	No deletion, collapse, branch selection, route selection, or closure admission is lawful without a locally generated certificate; a completed external trajectory is not a substitute.
Return-memory recurrence	Admitted closure returns through $\text{CIF}_s \rightarrow \text{QV}_s \rightarrow \text{RFL}_s \rightarrow M_{\text{rec}} \rightarrow \text{CIF}_{s+1}$.
Route-class evaluation	The operator acts on route classes, not arbitrary labels. Same closure is gauge equivalence; distinct valid closure is route strength.
No-loss quotienting	A representative may be removed only if all state, residual, invariant, projection, branch, return-memory, and kernel distinctions are preserved.
Public-data firewall	Public validation rows may not generate witnesses, tune constants, select routes, or modify normalization.
Collision event-lift	Exact collision is represented as an event boundary with source, collapse witness, rebirth envelope, return memory, and kernel action.
Kernel boundedness certificate	The bounded $\Phi_{j,[r],N}$ certificate supplies the majorant used in the convergence proof.

Theorem, Lemma, Proposition, and Protocol Inventory

Object	Reproduction role
Main theorem	States arbitrary finite- N body-family closure modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy.
Finite body system definition	Defines the finite Newtonian state object and admissible extension frame.
Directed lane definition	Defines $L(i \mid j)$ and the $N(N - 1)$ lane count.
Triadic packet definition	Defines CIF , QV , RFL , M_{rec} , K_f , First Action, and recurrence.
Certified admissibility definition	Lists the finite, positive-mass, time-normalized, represented-lane, locally witnessed, no-loss quotient, conservation, collision, and boundedness requirements.

Object	Reproduction role
Local witness generation protocol	Defines the differential residual and the certificate fields used to admit a candidate route.
Witness autonomy proposition	Establishes that route admission does not require a pre-solved global trajectory.
Lane growth lemma	Proves adding one body adds $2N$ directed lanes.
Add-one-body refinement lemma	Proves admissible one-body extension preserves old closure and may add lawful route classes.
Route-capacity proposition	Establishes added lawful routes as solution capacity rather than ambiguity.
Depth convergence lemma	Establishes absolute convergence of the depth series by geometric majorization.
Route-class normalization lemma	Establishes class-based coefficient invariance under representative relabeling.
Constructive closure operator proposition	Defines the kernel-series closure operator over admitted route classes.
Lawful branch selection protocol	Defines when a collision branch is uniquely selected and when a gauge-equivalent branch family is retained.
Collision branch symmetry proposition	Establishes tied lawful collision branches as gauge symmetry, not failure.

9 Proof Dependency Map

The proof can be reproduced from the following public theorem dependencies.

Dependency	Reproduction role
Directed lane construction	Define $L(i j)$ for every ordered pair $i \neq j$. Establish $N(N - 1)$ lanes.
Triadic packet definition	Attach CIF, QV, RFL, M_{rec} , and K_f to each admissible closure packet.
Time normalization	Declare a fixed nondimensionalization before route generation; recurrence order supplies memory ordering without a second clock.
Local witness generation	Construct differential-residual, invariant, projection, packet, branch, return-memory, and Φ certificates from the admitted state.
Closure predicate	Define when a route class lawfully closes a body-family packet.
Gauge rule	Same lawful closure under different representatives is gauge equivalence, subject to no-loss preservation of independent closure information.
Multi-route rule	Distinct lawful closures are route strength, not obstruction.
Add-one-body refinement	Adding one admissible body creates $2N$ new lanes and preserves or refines the closure class.
Collision event-lift	Exact collision is represented as a triadic event boundary with source, collapse witness, rebirth envelope, and return memory.

Dependency	Reproduction role
Lawful branch policy	Selects a branch only when the law distinguishes one; otherwise keeps the tied branch family as gauge symmetry.
Kernel convergence	Uses geometric depth damping and bounded triadic packet functions.
Finite-N assembly	Combines base case, add-one-body refinement, route equivalence, collision branch policy, and convergence.

10 Public Validation Source

The public numerical validation layer uses NASA/JPL Solar System Dynamics public interfaces.

Source	Public reference
API documentation	NASA/JPL Solar System Dynamics, Three-Body Periodic Orbits API, Version 1.0: https://ssd-api.jpl.nasa.gov/doc/periodic_orbits.html
API endpoint	https://ssd-api.jpl.nasa.gov/periodic_orbits.api
Interactive tool	NASA/JPL Solar System Dynamics, Three-Body Periodic Orbits: https://ssd.jpl.nasa.gov/tools/periodic_orbits.html
Fields used	x, y, z, v_x, v_y, v_z , Jacobi constant, period, stability

The author does not distribute private downloaded files. A reader can regenerate the public validation layer directly from the NASA/JPL API using the descriptor classes listed below. The API rows are read-only comparison objects: they do not create route witnesses, set constants, alter coefficients, or enter the route normalizer.

11 Public Query Descriptor Classes

The following descriptor classes specify the public validation layer. Query syntax follows the NASA/JPL API documentation. A descriptor with no libration point or branch leaves that parameter unset unless the API requires an additional family-specific option.

System	Family	Libration	Branch	Expected rows
Earth-Moon	axial	L1	none	2498
Earth-Moon	axial	L2	none	2198
Earth-Moon	halo	L1	S	5731
Earth-Moon	halo	L2	N	1535
Earth-Moon	halo	L2	S	1535
Earth-Moon	lyapunov	L1	none	3108
Earth-Moon	lyapunov	L2	none	4298
Earth-Moon	vertical	L1	none	6670
Earth-Moon	vertical	L2	none	9651
Jupiter-Europa	dpo	none	none	2388
Jupiter-Europa	dro	none	none	5767
Jupiter-Europa	halo	L1	N	819
Jupiter-Europa	halo	L1	S	819
Jupiter-Europa	halo	L2	N	860

System	Family	Libration	Branch	Expected rows
Jupiter-Europa	halo	L2	S	860
Saturn-Titan	butterfly	none	N	2558
Saturn-Titan	butterfly	none	S	2558
Total				53853

API Query Template

For a descriptor with system s , family f , libration point ℓ , and branch b , use the API pattern

`https://ssd-api.jpl.nasa.gov/periodic_orbits.api?sys=s&family=f&libr=l&branch=b.`

Omit `libr` and `branch` when the descriptor class has no such parameter. The API returns a JSON object containing system metadata, family information, row count, field names, and data rows.

12 Numerical Validation Protocol

The numerical validation layer is independent support for the public construction. It does not replace the proof.

12.1 Jacobi Recalculation

For each public CR3BP row with nondimensional state (x, y, z, v_x, v_y, v_z) and mass ratio μ , define

$$r_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2},$$

$$\Omega(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2},$$

and

$$C = 2\Omega(x, y, z) - (v_x^2 + v_y^2 + v_z^2).$$

The public validation target is agreement between recalculated C and the API-provided Jacobi value to numerical precision appropriate to the downloaded text precision and floating-point environment.

12.2 Propagation Check

For a selected public row, integrate the standard CR3BP equations over one API-provided period. The API model uses normalized CR3BP time, which is the same normalized-domain convention used in the preprint:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y}, \quad \ddot{z} = \frac{\partial \Omega}{\partial z}.$$

The return check compares the final state with the initial state in the infinity norm and separately records Jacobi drift. This validates that the public periodic-orbit states behave as periodic reference states under the CR3BP model.

13 Expected Public Validation Results

The following numbers are reproducibility targets from the public validation protocol. They are not private working materials and do not depend on private development archives.

Check	Expected public result
NASA/JPL rows extracted from descriptor classes	53853 rows
Jacobi recomputation	53853 / 53853 rows pass; maximum residual approximately 6.66×10^{-15} in the validation environment
Expanded one-period propagation sample	850 / 850 public rows pass
Maximum return infinity residual in sample	approximately 2.06×10^{-5}
Maximum Jacobi drift after one period in sample	approximately 4.49×10^{-10}
Route descriptors covered by propagation sample	17 / 17 descriptor classes

The full all-row propagation expansion is a public validation extension, not a condition for understanding the theorem statement. The preprint claim is mathematical and constructive; public numerical validation supports the reference layer.

14 Independent Proof Reproduction Procedure

A reader can reproduce the proof route without private files by following these steps.

1. Start from an arbitrary finite body set $\mathcal{B}_N$ with $N \geq 2$, declare the fixed time nondimensionalization, and define all directed lanes $L(i \mid j)$ for $i \neq j$.
2. Attach a triadic packet $(\text{CIF}, \text{QV}, \text{RFL}, M_{\text{rec}}, K_f)$ to each candidate closure route.
3. Construct local route jets and evaluate the directed-lane differential residual, invariant drift, projection consistency, packet continuity, collision status, branch policy, return memory, and Φ bound.
4. Admit only locally certified routes and define the closure predicate $\mathcal{C}_N([r])$ over lawful route classes.
5. Identify gauge-equivalent representatives as the same closure class only when no independent closure information is lost.
6. Identify distinct valid closure routes as multi-route strength, not contradiction.
7. Prove the add-one-body refinement step: an admissible new body adds $2N$ directed lanes and preserves or refines closure capacity.
8. Treat exact collision as an event-lift: pre-collision source, collapse witness, rebirth branch envelope, and return-memory continuation.
9. Apply the lawful branch policy: select a unique outgoing branch only when the law distinguishes one; otherwise retain the gauge-equivalent branch family.

10. Establish boundedness of $\Phi_{j,[r],N}$ and use the geometric depth factor $\delta^{-j}e^{-\alpha jt}$ to obtain constructive convergence.
11. Freeze the operator before public comparison, then assemble and independently validate the arbitrary finite- N closure theorem modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy.

15 Claim Boundaries

The claim boundaries are part of the public result.

Claim status	Meaning
Claimed	Constructive triadic closed-form kernel-series closure operator for arbitrary finite N under certified admissibility, modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy.
Claimed	More bodies create more directed lanes, and additional lawful closure routes are solution capacity rather than ambiguity.
Claimed	Public NASA/JPL CR3BP reference rows support the numerical validation protocol through Jacobi recomputation and expanded one-period propagation sampling.
Not claimed	Machine-checked formal proof in Lean, Coq, Isabelle, or another proof assistant.
Not claimed	Finite elementary coordinate formula for every body trajectory.
Not claimed	Arbitrary label-unique continuation through every exactly symmetric collision.
Not claimed	Peer-reviewed acceptance or community certification.
Not claimed	Distribution of private working materials or private development archives.
Not used	A completed NASA/JPL trajectory as a route-admission witness or source of fitted constants.
Not introduced	A second independent physical time coordinate; recurrence order organizes memory while t remains normalized domain time.

16 Public Bibliographic References

References

- [1] NASA Jet Propulsion Laboratory Solar System Dynamics Group. *Three-Body Periodic Orbits API*. Version 1.0, 2020. https://ssd-api.jpl.nasa.gov/doc/periodic_orbits.html. Accessed July 6, 2026.
- [2] NASA Jet Propulsion Laboratory Solar System Dynamics Group. *Three-Body Periodic Orbits*. Interactive Solar System Dynamics tool. https://ssd.jpl.nasa.gov/tools/periodic_orbits.html. Accessed July 6, 2026.
- [3] Szebehely, V. *Theory of Orbits: The Restricted Problem of Three Bodies*. Academic Press, 1967.
- [4] Meyer, K. R., Hall, G. R., and Offin, D. *Introduction to Hamiltonian Dynamical Systems and the N-Body Problem*. Springer, 2009.

A Minimal Public Reproduction Checklist

1. Read the preprint theorem, definitions, axioms, and witness protocol.
2. Declare and freeze the normalized-time convention.
3. Reconstruct the directed lane set $L(i \mid j)$ for finite N .
4. Implement local residual and witness certificates without importing solved trajectories.
5. Implement the triadic route-class kernel form and no-loss quotient rule.
6. Regenerate NASA/JPL public CR3BP rows only after freezing the operator.
7. Recalculate the Jacobi value for each row.
8. Propagate representative rows for one normalized period with the standard CR3BP equations.
9. Verify that route multiplicity is treated as lawful solution capacity when the closure conditions are met.
10. Preserve the stated claim boundaries and public-data firewall.

B Public Metadata Integrity Statement

This file intentionally contains only public-facing reproducibility content. It provides enough structure for a reader to understand the claim, reproduce the theorem route, regenerate the public validation layer, and evaluate the stated boundaries without access to the author’s private working files.